

Test 2: The Sun, the Atmosphere & Weather

Units 5–7 · pages 66–114 · 50 points · 50 minutes · Closed book

Name _____

Date _____

Score / 50

Read each question carefully. Circle one answer for multiple choice. Write on the lines provided; use the back of the page if you need more room. Point values are shown for each section.

Part A · Multiple Choice

12 questions × 2 points = 24 points

1. Which of the following does the text identify as a common *misconception* about the seasons?
 - a) That daylight lasts longer in summer than in winter
 - b) That seasons are caused by the tilt of Earth's axis
 - c) That summer is warm because the Sun is physically closer to Earth
 - d) That the Southern Hemisphere has summer when the Northern Hemisphere has winter
2. Ozone completely removes which type of ultraviolet radiation from incoming sunlight?
 - a) UVA
 - b) UVB
 - c) UVC
 - d) All three equally
3. Which greenhouse gas is *not* a normal part of Earth's atmosphere?
 - a) Carbon dioxide
 - b) Methane
 - c) Ozone
 - d) Chlorofluorocarbons (CFCs)
4. The ozone layer is located within which layer of the atmosphere?
 - a) Stratosphere
 - b) Mesosphere
 - c) Thermosphere
 - d) Troposphere
5. Which statement about the troposphere is correct?
 - a) It lies directly above the mesosphere
 - b) Temperature increases with altitude, making it very stable
 - c) Warm air sits beneath cooler air, making it unstable and causing constant mixing
 - d) It contains almost no water vapor

6. A candle flame and a bathtub of hot water are compared in the text. Which statement is correct?
- a) The flame has a higher temperature but less heat
 - b) The flame has both a higher temperature and more heat
 - c) The bathtub has a higher temperature but less heat
 - d) They have equal heat and equal temperature
7. The westerly winds of the Ferrel cell blow
- a) from the north toward the south
 - b) straight toward the equator
 - c) from the east toward the west
 - d) from the west toward the east
8. The polar front, where the polar jet stream is found, is the junction between
- a) the Hadley and Ferrel cells
 - b) the Ferrel and Polar cells
 - c) the two Hadley cells at the equator
 - d) the troposphere and mesosphere
9. Clouds form when air reaches its dew point. This can happen when
- a) temperature stays the same but humidity increases, or humidity stays the same but temperature decreases
 - b) temperature and humidity both decrease at the same rate
 - c) air pressure rises sharply
 - d) the air becomes drier
10. An air mass that is dry and cold would be classified as
- a) maritime tropical (mT)
 - b) continental tropical (cT)
 - c) maritime polar (mP)
 - d) continental polar (cP)
11. In a thunderstorm, the process by which downdrafts pull cold, dry air from outside the cloud toward the ground is called
- a) advection
 - b) entrainment
 - c) occlusion
 - d) convection
12. Which Köppen climate type is defined by evaporation exceeding precipitation, with a constant water deficiency throughout the year?
- a) Tropical (A)
 - b) Dry (B)
 - c) Temperate (C)
 - d) Continental (D)

Part B · Matching

1 question × 8 points = 8 points

13. Write the letter of the correct definition next to each term.

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|------------------------|--|
| _____ 1. Advection | a) The temperature at which saturated air condenses |
| _____ 2. Latent heat | b) A measure of how well a surface reflects light |
| _____ 3. Albedo | c) The ring of cloud holding a hurricane's strongest winds and lowest pressure |
| _____ 4. Dew point | d) The horizontal flow of air |
| _____ 5. Source region | e) A line of severe thunderstorms forming along a cold front |
| _____ 6. Squall line | f) Heat absorbed or released when a substance changes state |
| _____ 7. Nor'easter | g) A mid-latitude cyclone that strikes the mid-Atlantic and New England states |
| _____ 8. Eyewall | h) The area where an air mass acquires its temperature and humidity |

Part C · Short Answer

4 questions × 3 points = 12 points

14. Explain why the troposphere mixes constantly while the stratosphere barely mixes at all.

15. Explain the *rain shadow effect* and how it helps create deserts.

16. Name and briefly describe the three stages of a thunderstorm's life cycle.

17. Give two reasons the atmosphere makes life on Earth possible.

18. Trace the path of energy from the Sun to a hurricane. Explain how sunlight arriving in the tropics ultimately produces one of the most powerful storms on Earth. Your answer should refer to unequal heating of Earth's surface, evaporation and latent heat, low pressure, and the Coriolis effect.

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